

TOKMA, Russian Federation

WMO: 301270

Lat: 58.252N

Lon: 105.856E

Elev: 439

StdP: 96.16

Time Zone: 8.00 (E08)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -45.5	(c) -42.9	(d) -48.0	(e) 0.0	(f) -45.4	(g) -45.3	(h) 0.0	(i) -42.9	(j) 7.2	(k) -19.4	(l) 6.5	(m) -18.2	(n) 0.0	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	28.6	17.3	26.7	16.8	24.7	16.1	18.8	25.3	17.8	24.5	16.8	23.0	1.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.4	20.9	15.4	11.5	20.1	14.2	10.7	19.3	55.5	25.1	52.2	24.5	49.0	23.0	21.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.9	(b) 6.0	(c) 5.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 6.9	(b) 6.0	(c) 5.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-4.4	-25.4	-22.4	-11.7	-2.9	5.4	13.7	15.6	13.0	5.2	-5.0	-16.3	-23.4		
	DBStd	16.10	9.14	9.19	7.49	6.10	4.45	4.37	3.04	3.57	4.11	6.13	9.33	10.26		
	HDD10.0	5698	1098	908	674	387	156	16	1	14	154	466	789	1034		
	HDD18.3	8327	1356	1142	932	637	402	144	94	168	394	724	1039	1293		
	CDD10.0	432	0	0	0	0	13	128	174	107	9	0	0	0		
	CDD18.3	19	0	0	0	0	1	6	10	2	0	0	0	0		
	CDH23.3	688	0	0	0	0	9	302	244	127	6	0	0	0		
	CDH26.7	158	0	0	0	0	1	71	53	34	0	0	0	0		
	WSAvg	1.9	1.7	2.0	2.2	2.3	2.1	1.7	1.7	1.4	1.5	1.8	2.0	2.0		
	PrecAvg	389	18	13	13	21	33	51	62	55	38	34	29	23		
(15) Precipitation	PrecMax	475	39	24	29	41	58	88	118	97	74	55	48	42		
	PrecMin	300	6	3	5	9	13	18	16	15	12	11	16	8		
	PrecStd	44	7	4	6	7	11	16	27	19	13	12	8	9		
	PrecStd	44	7	4	6	7	11	16	27	19	13	12	8	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.0	-0.9	8.3	15.7	24.1	30.7	30.6	30.1	24.1	10.9	4.1	-0.9		
		MCWB	-2.4	-2.7	2.8	7.0	11.9	17.5	19.1	18.0	14.3	5.2	2.9	-1.9		
	2%	DB	-5.5	-3.2	4.5	12.2	20.5	28.7	28.3	26.9	19.6	7.5	1.3	-4.9		
		MCWB	-6.1	-4.2	0.5	5.2	10.2	16.7	18.0	17.1	12.0	3.8	0.7	-5.5		
	5%	DB	-8.1	-5.9	1.6	9.6	17.9	27.1	26.2	23.9	16.2	4.8	-1.1	-6.3		
		MCWB	-8.8	-7.0	-1.3	3.9	9.2	16.5	17.2	15.4	10.4	2.2	-1.9	-6.9		
	10%	DB	-11.1	-8.7	-0.8	6.8	15.1	24.9	24.2	21.4	13.2	2.7	-3.5	-8.5		
		MCWB	-11.6	-9.6	-3.0	2.3	7.8	15.6	16.4	14.9	8.7	0.9	-4.4	-9.0		
	0.4%	WB	-2.9	-2.7	3.5	7.2	13.3	19.4	20.3	19.2	14.8	6.1	3.5	-1.9		
		MCDB	-2.6	-1.5	8.0	13.6	22.8	26.3	27.0	25.5	22.0	9.7	4.2	-1.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-6.0	-4.3	0.8	5.6	11.4	18.2	19.2	18.1	13.0	4.1	0.6	-5.5		
		MCDB	-5.4	-3.3	4.3	11.5	17.7	26.4	25.4	24.4	18.4	7.0	1.2	-4.9		
	5%	WB	-8.7	-7.0	-1.2	4.1	9.9	17.2	18.2	17.0	11.0	2.5	-1.9	-6.9		
		MCDB	-8.2	-6.0	1.2	8.8	16.4	25.0	24.4	22.4	15.4	4.5	-1.1	-6.3		
	10%	WB	-11.5	-9.5	-2.9	2.6	8.4	16.1	17.1	15.7	9.3	1.0	-4.3	-8.9		
		MCDB	-11.0	-8.7	-0.8	6.8	14.3	23.5	22.9	20.1	12.6	2.5	-3.5	-8.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	14.7	16.0	13.1	13.0	15.8	14.0	14.6	12.8	9.7	10.3	10.2		
		MCDBR	11.2	11.2	17.2	16.0	19.1	20.8	18.9	21.0	19.4	13.2	8.3	8.8		
		MCWBR	11.0	10.1	13.6	10.0	10.4	10.0	9.6	11.3	11.4	9.5	7.8	8.3		
		MCDBR	11.2	11.3	15.9	13.9	15.4	17.8	15.6	16.5	18.0	12.0	8.3	8.6		
	5% WB	MCWBR	10.9	10.3	12.8	9.1	9.0	8.9	8.6	9.5	11.1	8.9	8.0	8.2		
		MCWBR	10.9	10.3	12.8	9.1	9.0	8.9	8.6	9.5	11.1	8.9	8.0	8.2		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.293	0.315	0.347	0.377	0.418	0.482	0.410	0.335	0.312	0.268	0.242		
	taud		1.998	2.022	2.061	2.141	2.288	2.254	2.054	2.253	2.449	2.314	2.056	1.954		
	Ebn at Noon		541	709	814	855	854	819	747	785	807	704	528	428		
	Edh at Noon		56	90	116	127	118	125	150	114	80	68	52	42		
(44) All-Sky Solar Radiation	RadAvg		0.64	1.60	3.08	4.50	5.13	5.71	5.21	4.18	2.59	1.27	0.67	0.38		
	RadStd		0.06	0.11	0.19	0.27	0.34	0.44	0.44	0.31	0.27	0.14	0.08	0.05		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page